

AI-DRIVEN VISION BASED INVENTORY MANAGEMENT SYSTEM

A PROJECT REPORT

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AFZAL K S L

(2024179033)

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DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING, GUINDY

ANNA UNIVERSITY

CHENNAI 600 025

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ANNA UNIVERSITY

CHENNAI - 600 025

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PLACE: CHENNAI

DATE:

Dr. M. DEIVAMANI

ASSISTANT PROFESSOR

PROJECT GUIDE

DEPARTMENT OF IST, CEG

ANNA UNIVERSITY

CHENNAI 600 025

COUNTERSIGNED

DR. S. SWAMYNATHAN

HEAD OF THE DEPARTMENT

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING, GUINDY

ANNA UNIVERSITY

CHENNAI 600025

ABSTRACT

This project proposes an AI-driven vision-based inventory management system designed to eliminate stock-out situations and improve operational efficiency through intelligent shelf monitoring. By leveraging a video feed and smart vision capture triggered by human presence, the system continuously monitors shelf activity. It utilizes cameras and deep learning models for real-time product detection, counting, and shelf monitoring without physical human intervention.

The core of the system is a dual-layered analytical engine designed to handle the complexities of dense retail environments. First, an advanced 2D object detection model—optimized through transfer learning—identifies visible front-facing products and assigns bounding boxes to classify current stock levels. Second, to overcome the limitations of standard 2D systems, the architecture integrates a monocular depth estimation model. By generating depth maps and calculating relative physical scale, the system actively addresses occlusion and overlapping items, allowing it to accurately estimate hidden, stacked inventory behind the front row.

The output is a seamless transition from detection to action, once the total calculated stock drops below a predefined threshold, the system automatically generates restocking decisions. To ensure human oversight, the architecture includes a web-based interface for inventory visualization and control. This dashboard provides annotated images, front counts, and total estimated inventory, significantly reducing manual workload while maintaining a hybrid control environment between autonomous AI operations and human management.

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Chapter 1

INTRODUCTION

1.1 OVERVIEW

Retail inventory management is a critical operational process that directly impacts customer satisfaction, business profitability, and supply chain efficiency. Traditional inventory management systems rely heavily on manual audits, barcode scanning, and periodic stock verification, which are labor-intensive, error-prone, and inefficient for large-scale retail environments. In fast-moving consumer goods (FMCG) retail, stock-out and overstock situations frequently occur due to delayed shelf monitoring and inaccurate stock estimation.

With the advancement of computer vision and artificial intelligence, automated visual monitoring systems have emerged as a promising solution for intelligent retail environments. Vision-based systems using cameras and deep learning models enable real-time product detection, counting, and shelf monitoring without physical human intervention. Recent developments in deep learning-based object detection and monocular depth estimation models have further enhanced the ability to understand complex retail shelf environments.

This project proposes an AI-driven vision-based inventory management system that integrates real-time image capture, deep learning-based product detection, depth estimation, automated stock analysis, and web-based inventory visualization. The system aims to provide intelligent shelf monitoring and automated restocking decisions to improve inventory accuracy and operational efficiency.

1.2 PROBLEM STATEMENT

Existing inventory management systems suffer from several limitations:

- Manual shelf inspection and stock verification
- Inaccurate product counting due to occlusion and overlapping items
- Delayed detection of low-stock and out-of-stock situations
- Lack of real-time monitoring and automation

- High operational cost and human dependency

These challenges lead to stock-out conditions, revenue loss, and reduced customer satisfaction. Therefore, there is a need for an intelligent, automated, and vision-based inventory monitoring system that can detect products, estimate quantities, handle occlusions, and support real-time decision-making.

1.3 MOTIVATION

The motivation for this project arises from the increasing demand for automation and intelligence in retail systems. With the growth of smart retail and AI-driven automation, there is a strong need for systems that can autonomously monitor shelves, analyze product availability, and assist inventory decision-making without continuous human intervention.

This project is motivated by the potential of computer vision, deep learning, and depth estimation models to transform traditional inventory systems into intelligent systems. By combining object detection, depth-based analysis, and automated decision logic, the proposed system aims to reduce human workload, improve accuracy, and enhance retail operational efficiency.

1.4 OBJECTIVES

The main objectives of the project are:

- To design an AI-driven vision-based inventory management system
- To implement automated image capture using human presence detection
- To detect and classify products using deep learning models
- To estimate depth information for occlusion-aware counting
- To perform automated inventory threshold analysis
- To generate restocking decisions based on stock availability
- To develop a web-based interface for inventory visualization and control
- To integrate database storage for inventory records

1.5 SCOPE OF THE PROJECT

The scope of the project includes:

- Automated shelf image acquisition
- Vision-based product detection and counting
- Depth-based occlusion handling
- Inventory decision automation
- Web interface and database integration

At the current implementation stage, the system focuses on video processing based on person presence and detection-based counting. Future enhancements include metric depth estimation, predictive inventory analytics and web interface.

Chapter 2

LITERATURE SURVEY

This chapter presents a review of existing research works related to intelligent retail systems, automated inventory monitoring, computer vision-based shelf analysis, and depth-based product availability estimation. The literature highlights the evolution of retail automation technologies and identifies the research gap addressed by the proposed system.

2.1 EXISTING WORK

Traditional retail inventory management systems primarily rely on manual shelf audits, barcode scanning, and periodic stock verification. These methods are labour-intensive, time-consuming, and prone to human error, especially in large-scale retail environments. As a result, stock-out situations often remain undetected until customer demand is affected [1].

To overcome these challenges, several automated approaches have been explored. Early solutions employed RFID-based inventory tracking, where tagged products are monitored using radio-frequency readers. While RFID improves inventory visibility, it involves high implementation cost, item-level tagging, and infrastructure complexity [4]. Similarly, weight sensor-based smart shelves have been proposed to infer product availability by measuring changes in shelf weight. However, these systems lack product identification capability and are sensitive to product placement variations [2].

Infrared and proximity sensor-based systems have also been studied for shelf monitoring. Although they provide basic stock presence information, they fail to capture detailed shelf conditions and product-level insights. Due to these limitations, researchers have increasingly focused on vision-based approaches for automated inventory monitoring [5].

2.2 INTELLIGENT RETAIL AND AUTOMATED SHELF MONITORING

Automated shelf monitoring has gained significant attention due to the limitations of manual inventory audits. Early approaches relied on RFID systems, weight sensors,

and infrared-based sensing techniques. Although these systems improved stock visibility, they introduced high installation costs and infrastructure complexity [4][2].

Recent advancements in computer vision have enabled camera-based shelf monitoring systems that use image processing and machine learning techniques for detecting products and estimating availability. Vision-based systems provide scalable and cost-effective alternatives for intelligent retail automation and reduce dependency on specialized hardware [5].

2.3 DEPTH-BASED SHELF ANALYSIS IN RETAIL SYSTEMS

Milella et al. proposed an automated on-shelf availability estimation framework using a depth camera [1]. Their system utilizes RGB-D sensors to reconstruct 3D point clouds of shelves and estimate product availability based on surface reconstruction and occupancy grid modeling. The system compares a reference shelf model with current shelf states to detect stock variations.

The study demonstrates that depth sensing enables improved estimation of product availability by addressing occlusion and overlapping challenges common in retail shelves. The proposed method does not require prior product knowledge and achieves high accuracy in real retail environments[1].

This work forms a foundational reference for depth-based retail inventory analysis and motivates the integration of depth estimation models in vision-based inventory systems.

2.4 VISION-BASED INVENTORY SYSTEMS USING DEEP LEARNING

Modern retail systems increasingly adopt deep learning models such as convolutional neural networks (CNNs) for product detection, classification, and shelf analysis. Object detection models including YOLO, Faster R-CNN, and SSD have demonstrated high accuracy in detecting products from shelf images [5][7].

Deep learning-based systems provide robustness to lighting variations, camera angles, and shelf layouts. However, most 2D vision-based systems struggle with occlusion, depth ambiguity, and overlapping products, which limits counting accuracy in dense shelf environments [5][1].

2.5 OBSERVATIONS FROM EXISTING WORK

Based on the literature survey, the following observations can be made:

- Automated shelf monitoring is essential for reducing manual effort and improving inventory accuracy.
- Sensor-based solutions such as RFID and weight sensors improve stock visibility but suffer from high cost and limited flexibility.
- Vision-based systems provide richer contextual information and better scalability than traditional sensor-based approaches.
- Deep learning models significantly improve product detection and classification accuracy.
- Most existing vision-based systems operate on 2D image data, making them vulnerable to occlusion and overlapping products.
- Depth-based approaches improve shelf understanding but introduce challenges related to sensor noise and environmental variability.

These observations indicate that while substantial progress has been made, existing systems still face practical challenges in real-world retail environments.

2.6 LIMITATIONS OF EXISTING SYSTEMS

Despite advancements in automated inventory monitoring, current systems exhibit several limitations:

- Sensitivity to occlusion caused by overlapping products and shopper presence [5][1].
- Depth estimation inaccuracies due to sensor noise and environmental conditions [1].
- Dependency on lighting conditions in RGB-based vision systems.
- Limited integration between detection, depth analysis, and inventory decision logic.
- High infrastructure cost or calibration requirements in sensor-heavy systems [2].

2.7 RESEARCH GAP

From the literature review, it is evident that there is a lack of integrated inventory management systems that combine automated image capture, vision-based product detection, depth-aware shelf analysis, and inventory decision support within a unified framework.

Most existing solutions address individual components of the problem, such as detection or availability estimation, but do not provide an end-to-end solution suitable for real-time retail deployment. Additionally, robust handling of occlusion and overlapping products remains a critical challenge in vision-based inventory systems [1][5].

2.8 SUMMARY OF LITERATURE SURVEY

This chapter reviewed existing research on automated shelf monitoring and intelligent retail systems. Traditional sensor-based methods, vision-based approaches, and depth-based shelf analysis techniques were discussed along with their strengths and limitations. While vision and depth sensing have significantly improved inventory monitoring, challenges such as occlusion, depth noise, and system integration persist.

The identified research gap highlights the need for an AI-driven, vision-based inventory management system that integrates detection, depth analysis, and inventory intelligence in a unified framework.

Chapter 3

SYSTEM ARCHITECTURE

The system architecture of the Smart Retail Inventory Management System is designed as a modular, end-to-end pipeline that transitions from live video input through intelligent frame capture, deep learning-based product detection, monocular depth estimation, and automated restocking logic to a final inventory status accessible through a web-based interface. The architecture ensures that every shelf image is systematically captured, preprocessed, analysed, and evaluated before a restocking decision is recorded.

3.1 HIGH-LEVEL ARCHITECTURAL DESIGN

The Smart Retail Inventory Management System is structured into four primary functional modules and one interface layer that collectively manage the complete lifecycle of a shelf inventory monitoring cycle:

1. **Smart Vision Capture (Module 1):** Monitors live video for human presence within a defined shelf region of interest and triggers image capture only during verified occlusion-free windows.
2. **Product Detection and Front Count (Module 2):** Applies data preprocessing and a fine-tuned YOLOv8l object detection model to the captured image to identify all visible products and produce a class-wise front count.
3. **Stacked Product Estimation (Module 3):** Processes bounding box output through Depth Anything V2 to generate depth maps, estimate stack counts using empirical calibration and K-Means clustering, and compute total per-product inventory.
4. **Restock and Threshold Monitoring (Module 4):** Evaluates final inventory counts against predefined minimum thresholds and generates restock records persisted to a central database.
5. **Web User Interface:** Provides a browser-based interface displaying live video, annotated detection output, an inventory dashboard, and a restock management panel for store operators.

3.2 OVERALL SYSTEM ARCHITECTURE

The proposed smart retail inventory management system is designed as a modular, end-to-end pipeline that leverages computer vision, deep learning, and depth estimation to automatically monitor product availability on store shelves. The system operates in real time through a web application and requires no physical sensors or manual intervention for standard inventory tracking.

The architecture is composed of six primary modules that work in sequence: Smart Vision Capture, Product Detection, Depth Estimation, Restocking, and a Web User Interface. Each module is responsible for a distinct processing stage, and together they form an integrated pipeline from raw video input to actionable inventory data stored in a database.

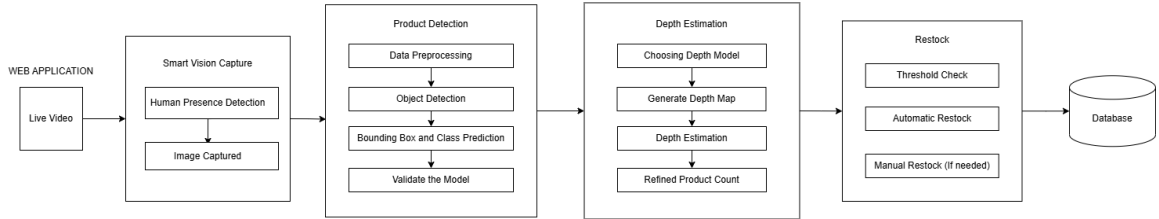


Figure 3.1: Overall System Architecture of the AI Driven Vision Based Inventory Management System

The system begins with a live video feed from the web application. The Smart Vision Capture module detects human presence near a shelf and triggers image capture. The captured image then enters the Product Detection module, where preprocessing, object detection, and model validation are performed. The detected products are passed to the Depth Estimation module, which generates a depth map and refines the product count to account for stacked items. Finally, the Restock module compares inventory levels against thresholds and triggers automatic or manual restocking actions, persisting all records to a central database.

3.3 FRAME EXTRACTION AND FEATURE EXTRACTION

3.3.1 SMART VISION CAPTURE

The system ingests a continuous live video stream delivered through the web application interface. Rather than processing every frame, which would impose significant computational overhead, the system employs an intelligent frame extraction strategy based on human presence detection.

When a customer approaches the monitored shelf, a presence detection algorithm identifies motion or human proximity within the camera's field of view. This event serves as the trigger for image capture. The precise moment of capture is selected to minimize motion blur and maximize shelf visibility, ensuring the captured frame is of sufficient quality for downstream detection.

3.3.2 FEATURE EXTRACTION VIA YOLOV8 BACKBONE

Once an image is captured, it is passed through the YOLOv8l model, which employs a convolutional neural network backbone to extract hierarchical spatial features from the input image. The backbone progressively encodes low-level texture features (edges, colors) in early layers and high-level semantic features (product shapes, labels) in deeper layers.

These extracted feature maps are then passed to the model's neck (a Feature Pyramid Network) and detection head, which are responsible for predicting bounding boxes, confidence scores, and class labels. The use of a large YOLOv8l model ensures that dense shelf arrangements with tightly packed products are handled accurately.

3.4 DATA PREPROCESSING

Before a captured image is fed into the detection model, it undergoes a series of preprocessing steps to standardize its format and optimize model performance.

3.4.1 IMAGE RESIZING AND LETTERBOXING

All input images are resized to a fixed resolution of 640×640 pixels using a letterboxing approach. Unlike simple stretching, letterboxing preserves the original aspect ratio by padding the shorter dimension with gray borders. This is critical because stretching distorts the geometric proportions of product packaging, which would degrade bounding box accuracy and class prediction.

3.4.2 AUTO-ORIENTATION

Images captured from varied camera angles or devices may contain EXIF orientation metadata that causes misalignment during processing. The auto-orientation step reads and applies this metadata, ensuring all images are correctly upright before entering the model pipeline.

3.4.3 DATA AUGMENTATION (TRAINING PHASE)

During the model training phase, augmentation techniques are applied to increase dataset diversity and model robustness. The following augmentations are employed:

- **Horizontal Flip:** Simulates products placed from different orientations.
- **Rotation (-10 to $+10$):** Accounts for slight camera tilt variations.
- **Brightness Adjustment ($\pm 20\%$):** Handles varying store lighting conditions.
- **Gaussian Blur (up to 1.5px):** Improves robustness to minor camera defocus.
- **Bounding Box Cutout:** Forces the model to recognize partially occluded products, which is common in dense shelf environments.

These augmentations are especially important given the relatively small size of the custom Reliance dataset. By artificially expanding the effective training distribution, the model learns to generalize to real-world shelf conditions.

3.5 DETECTION PROCESS

3.5.1 TRANSFER LEARNING STRATEGY

The detection model is built using a two-stage transfer learning approach. In the first stage, YOLOv8l is pre-trained on the SKU-110k dataset, a large-scale benchmark containing over 110,000 images of densely packed retail shelf products. This stage teaches the model the visual grammar of retail shelves: how to separate tightly adjacent products, identify shelf edges, and handle overlapping bounding boxes.

In the second stage, the pre-trained weights (saved as sku110k_base.pt) are loaded as a starting point, and the model is fine-tuned on the custom Reliance dataset containing approximately 200 product classes specific to the target retail environment. During fine-tuning, the backbone retains its learned shelf geometry representations while the detection head adapts to the specific product classes in scope.

3.5.2 OBJECT DETECTION AND BOUNDING BOX PREDICTION

During inference, the preprocessed image is passed through the fine-tuned YOLOv8l model. The model simultaneously predicts:

- Bounding box coordinates (x, y, width, height) for each detected product.
- A class label identifying the product type from the 200 defined categories.
- A confidence score indicating the model's certainty for each prediction.

Predictions below a defined confidence threshold are discarded to reduce false positives, particularly important in dense shelf scenarios where background clutter may trigger spurious detections.

3.5.3 NON-MAXIMUM SUPPRESSION (NMS) TUNING

A key challenge in dense product detection is that multiple overlapping bounding boxes may be predicted for a single item. Non-Maximum Suppression (NMS) is applied to eliminate redundant detections by retaining only the highest-confidence box when overlap between boxes exceeds a set Intersection over Union (IoU) threshold.

For standard object detection tasks, an IoU threshold of 0.45 is common. However, because shelf products are tightly packed and legitimately have high spatial overlap with adjacent items, the IoU threshold is raised to 0.6–0.7. This ensures that valid detections for neighbouring products are preserved rather than being suppressed incorrectly.

3.5.4 MODEL VALIDATION

The detection pipeline includes a validation stage to evaluate model performance on a held-out test set (10% of the dataset). Key metrics monitored during validation include:

- **Mean Average Precision (mAP@0.5):** Primary benchmark for detection accuracy.
- **Precision and Recall:** Measures of false positive and false negative rates.
- **Confusion Matrix:** Identifies class-level misclassification patterns.

Validation results guide iterative improvements to the model, including class filtering (removing classes with fewer than 5 training samples) and augmentation strategy adjustments.

3.6 DEPTH ESTIMATION

Front-facing product detection alone captures only the visible layer of products on a shelf. To estimate total stock including products stacked behind the front row, the system incorporates a depth estimation module.

3.6.1 DEPTH MODEL SELECTION

The system uses Depth Anything V2, a state-of-the-art monocular depth estimation model. Depth Anything V2 operates on a single RGB image and produces a dense relative depth map, assigning a depth value to every pixel in the image. Unlike stereo cameras or LiDAR sensors, this approach requires no additional hardware, making it suitable for deployment with standard web cameras.

It is important to note that the depth values produced are relative rather than absolute. The scale of depth values varies between images and does not directly represent distances in centimetres without calibration.

3.6.2 REFERENCE OBJECT CALIBRATION

To convert relative depth values into physically meaningful measurements, the system performs reference object calibration. A product of known physical width (e.g., a standard 15 cm product) is identified within the detected bounding boxes. The pixel width of this product's bounding box is measured from the detection output.

The physical-to-pixel scale factor is then calculated as follows:

$$\text{Scale (cm/pixel)} = \frac{\text{Known Physical Width (cm)}}{\text{Bounding Box Width (pixels)}} \quad (3.1)$$

This scale factor is applied to depth measurements, enabling the system to estimate real-world distances within the scene without requiring camera intrinsic parameters or calibration targets.

3.6.3 STACK COUNT ESTIMATION

Once depth calibration is established, the stack count for each detected product column is estimated using the following procedure:

- The relative depth of the front product is extracted from the depth map at the bounding box location.

- The relative depth of the shelf back wall is identified (either from the depth map at the boundary of detected products or from a reference depth established during setup).
- The depth gap between the front product and the back wall is calculated.
- The stack count is estimated by dividing this gap by the approximate depth of a single product unit.

$$\text{Stack Count} = \frac{\text{Depth Gap}}{\text{Product Depth}} \quad (3.2)$$

The total estimated inventory for each product is then computed as the product of the front count from YOLO detections and the estimated stack count from depth analysis.

3.6.4 HANDLING OCCLUSION AND EMPTY SHELF DETECTION

A known limitation of front-facing cameras is the inability to directly observe products behind the front row. Two strategies are employed to address this:

- **Camera Angle Adjustment:** Positioning the camera at a slight downward angle of 20–30 degrees allows the depth map to capture the shelf surface and back wall, improving the accuracy of gap estimation.
- **Empty Shelf Class Training:** A dedicated Empty_Shelf class is included in the detection model’s training data. Annotated samples of empty shelf sections allow YOLO to directly detect stock-out regions, providing an explicit signal for inventory depletion beyond what depth estimation alone can infer.

3.7 THRESHOLD CHECK AND RESTOCKING LOGIC

3.7.1 INVENTORY THRESHOLD DEFINITION

Each product class in the system is associated with a minimum inventory threshold, defined by store managers during system configuration. This threshold represents the minimum acceptable product count on the shelf below which restocking action is required. Thresholds can be configured differently for high-demand, perishable, or promotional products.

3.7.2 THRESHOLD EVALUATION

After each detection and depth estimation cycle, the refined product count for every identified SKU is compared against its predefined threshold. If the estimated count falls at or below the threshold, the system flags that product for restocking. This comparison runs automatically after every captured image is processed, enabling near-real-time inventory monitoring.

3.7.3 AUTOMATIC RESTOCKING

For products identified as requiring restocking, the system automatically generates a restock order entry in the connected database. This record includes the product identifier, current estimated count, required quantity to reach optimal stock levels, and the timestamp of the detection event. In integrated retail environments, this database entry can be consumed by supply chain management systems to trigger procurement or warehouse fulfillment workflows.

3.7.4 MANUAL RESTOCKING OVERRIDE

The system also supports a manual restocking pathway for situations where automated decisions require human confirmation, such as when detection confidence is low, products are irregularly arranged, or promotional stock rules apply. Store staff can review flagged items through the Web UI and confirm, modify, or dismiss restock orders as appropriate.

3.8 WEB USER INTERFACE

The Web User Interface serves as the primary interaction layer between the system and store operators. It is built as a browser-based application accessible from both desktop and mobile devices, enabling flexible monitoring across store environments.

3.8.1 LIVE VIDEO AND CAPTURE FEED

The UI displays a live video feed from the shelf camera. When the Smart Vision Capture module triggers an image capture event, the corresponding captured frame is displayed alongside the live feed, providing operators with immediate visual confirmation of what the system has processed.

3.8.2 ANNOTATED DETECTION OUTPUT

Following inference, the UI renders the captured image with overlaid bounding boxes drawn around each detected product. Each bounding box is labelled with the predicted product class name and confidence score. Products flagged as low-stock are visually highlighted (e.g., with a distinct border colour) to immediately draw operator attention.

3.8.3 INVENTORY DASHBOARD

The dashboard view presents a tabular and graphical summary of current inventory levels across all monitored product categories. For each SKU, the display includes:

- Current estimated front count (from YOLO detection).
- Estimated stack count (from depth estimation).
- Total estimated inventory.
- Threshold status (Normal / Low / Out-of-Stock).
- Last updated timestamp.

3.8.4 RESTOCK MANAGEMENT PANEL

A dedicated restock panel lists all products currently flagged for restocking. Operators can view automatically generated restock recommendations, adjust quantities if needed, confirm automatic orders, or escalate items for manual restocking. All actions taken through this panel are logged with user credentials and timestamps, providing a full audit trail.

3.8.5 DATABASE INTEGRATION

All inventory events, detection results, depth estimates, and restock actions are persisted to a central database. The Web UI provides read and write access to this database, enabling historical analysis, trend monitoring, and reporting. The database schema is structured around product identifiers, timestamp-based inventory snapshots, and restock event records, supporting both operational queries and analytical use cases.

3.9 SUMMARY

This chapter described the complete system architecture of the proposed smart inventory management solution. The pipeline integrates Smart Vision Capture for intelligent image acquisition, a transfer learning-based YOLOv8 detection module for product identification, Depth Anything V2 for monocular depth estimation and stack counting, and an automated threshold-driven restocking engine. All components are coordinated through a responsive web interface that provides real-time visibility and control to store operators. The following chapter will detail the implementation methodology and experimental setup used to train, validate, and deploy the described system components.

Chapter 4

MODULE DESCRIPTION

This chapter provides a detailed description of each module that constitutes the smart retail inventory management system. The system is composed of four functional modules and one interface module, each designed to perform a well-defined role in the end-to-end pipeline. The modules operate sequentially, passing structured outputs from one stage to the next, culminating in a real-time inventory status accessible through a web-based interface.

4.1 SMART VISION CAPTURE MODULE

The Smart Frame Capture module is responsible for autonomously acquiring shelf images at the right moment specifically when the shelf is free of human occlusion. A continuous video stream from a CCTV camera or an uploaded video file serves as the raw input. Rather than blindly sampling every frame, the module applies intelligent filtering to ensure that only frames showing a clear, unobstructed shelf are selected for downstream analysis.

The module begins by defining a Region of Interest (ROI) that corresponds to the shelf area within the camera's field of view. This boundary is configured once during system setup and ensures that human presence detection is restricted to the shelf zone, ignoring pedestrians passing in adjacent areas. Each incoming video frame is evaluated against this ROI using a YOLOv8 model loaded with standard COCO weights, which detects persons as class 0. If any person bounding box intersects with the ROI, the system suppresses capture and continues monitoring subsequent frames.

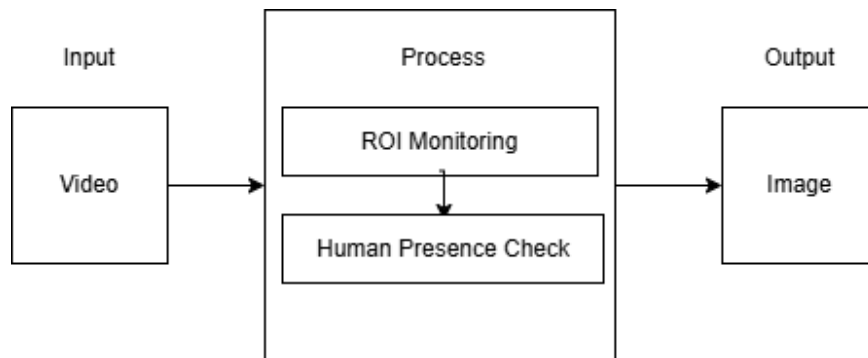


Figure 4.1: Smart Vision Capture Module: ROI Monitoring and Human Presence Check

A key design feature of this module is the application of temporal logic. Rather than capturing the very first frame after a person exits the ROI, the module waits for a continuous, uninterrupted 3-second window during which no person is detected within the shelf region. This threshold prevents captures during brief pauses in customer movement and ensures that the shelf is genuinely clear before a snapshot is taken. Once the condition is satisfied, the current frame is captured and saved as a high-resolution image, along with metadata including the capture timestamp and camera details. This clean snapshot is then passed as input to Module 2.

The rationale for this design is straightforward: product detection and depth estimation are highly sensitive to occlusions. When a person is standing in front of the shelf, products are hidden behind their body and the depth map generated in Module 3 would reflect the person’s silhouette rather than the shelf surface. By ensuring captures only occur on a clear shelf, Module 1 eliminates a major source of detection error at its root.

4.2 PRODUCT DETECTION AND FRONT COUNT MODULE

The Product Detection module takes the clean shelf image produced by Module 1 and applies a fine-tuned object detection model to identify all visible products and produce a class-wise front count. The front count refers to the number of product units that are directly visible in the front row of the shelf — the layer immediately facing the camera.

Before inference, the image is preprocessed to meet the model’s input requirements. This includes letterbox resizing to 640×640 pixels, which scales the image to the target resolution while preserving the original aspect ratio by padding shorter dimensions with grey borders. Preserving aspect ratio is critical because distorting product geometry through simple stretching degrades the accuracy of bounding box predictions. Auto-orientation correction is also applied to handle images captured from devices that embed rotation metadata in the file header.

Detection is performed using a YOLOv8l model trained through a two-stage transfer learning process. In the first stage, the model is pre-trained on the SKU-110k dataset, a large-scale benchmark containing over 110,000 annotated images of densely packed retail shelves. This pre-training phase allows the model backbone to develop strong representations of shelf geometry, tight product boundaries, and edge separation between adjacent items. In the second stage, these learned weights are used as a starting point for fine-tuning on the custom product dataset, which

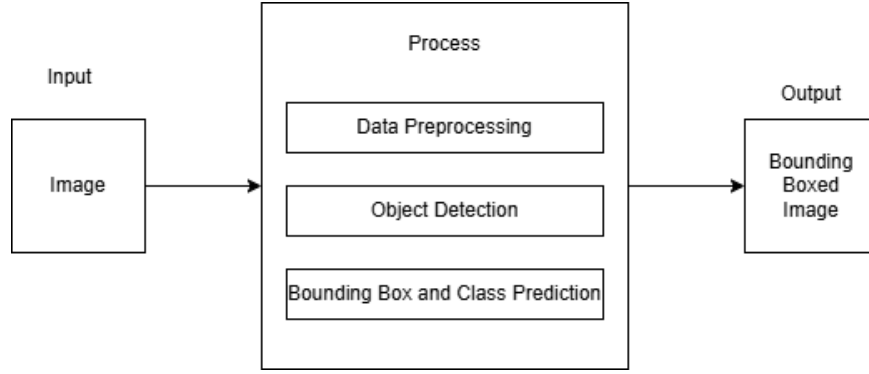


Figure 4.2: Product Detection Module: YOLOv8 Inference and Bounding Box Prediction

contains approximately 200 product classes specific to the target retail environment. The detection head, which is responsible for predicting class labels, is retrained to recognise these specific classes while the backbone retains its shelf-domain knowledge.

During inference, the model simultaneously predicts bounding box coordinates, class labels, and confidence scores for all detected products in the image. Predictions below a minimum confidence threshold are discarded. Non-Maximum Suppression (NMS) is then applied to resolve duplicate detections caused by overlapping predictions for the same product. Because shelf products are legitimately adjacent with high spatial overlap, the IoU threshold for NMS is raised to 0.6–0.7 to prevent valid detections for neighbouring products from being incorrectly suppressed.

The final output of this module is a bounding-boxed annotated image with class labels overlaid, a class-wise front count table showing the number of visible units per product, and a structured detection summary containing bounding box coordinates and confidence scores for each detected item. The annotated image is displayed in the Web UI, and the detection summary is passed to Module 3 to guide depth extraction.

4.3 STACKED PRODUCT ESTIMATION

The Stacked Product Estimation module extends the inventory count beyond what is directly visible to the camera. In a typical retail shelf, identical products are stacked in depth behind the front-facing unit, meaning that a front count alone significantly underestimates the actual stock level. This module addresses that limitation by applying monocular depth estimation to infer the number of hidden units stacked behind each detected product.

The bounding-boxed image from Module 2 is passed through Depth Anything V2, a state-of-the-art monocular depth estimation model that operates on a single RGB image and produces a dense inverse depth map. In this map, each pixel is assigned a relative depth value where higher values correspond to surfaces closer to the camera. The model requires no specialised hardware such as stereo cameras or depth sensors, making it deployable with standard shelf-mounted web cameras. It is important to note that the depth values are expressed in arbitrary relative units, and their absolute scale differs between images depending on the scene content and camera positioning.

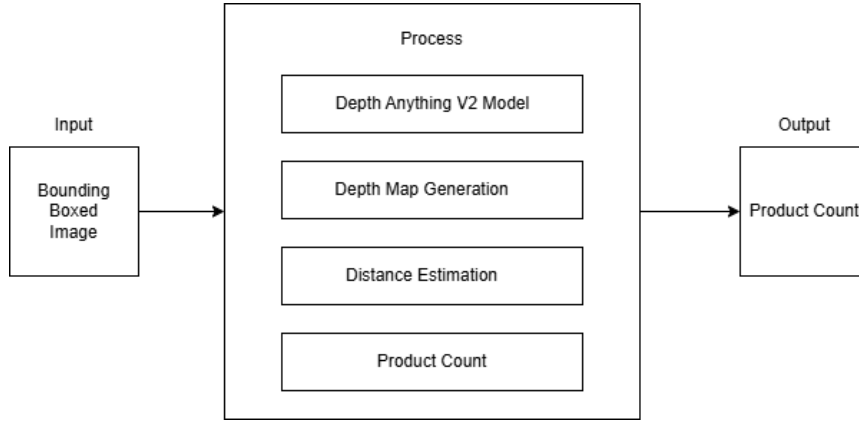


Figure 4.3: Stacked Product Estimation Module: Depth Map Generation and Stack Count Estimation

To translate these relative depth values into physically meaningful distances, an empirical calibration step is performed for each bounding box received from Module 2. The median depth of pixels within the bounding box interior is taken as the front product depth, representing the surface of the visible product. A narrow reference strip sampled just above the top edge of the bounding box provides the back wall depth, representing the shelf’s rear surface or background. The absolute depth gap between these two measurements is computed as:

$$\text{depth_gap} = |\text{front_depth} - \text{wall_depth}| \quad (4.1)$$

A calibrated constant (`DEPTH_UNITS_PER_CM`), determined empirically during system setup by comparing the known physical depth of a reference product against its measured depth gap, is used to convert the relative gap into centimetres:

$$\text{physical_gap_cm} = \frac{\text{depth_gap}}{\text{DEPTH_UNITS_PER_CM}} \quad (4.2)$$

The number of hidden products stacked behind the front unit is then estimated by dividing this physical gap by the known depth of a single product unit (`product_depth_cm`), a class-specific constant:

$$\text{estimated_extra} = \frac{\text{physical_gap_cm}}{\text{product_depth_cm}}, \quad \text{stack_gap} = 1 + \lfloor \text{estimated_extra} \rfloor \quad (4.3)$$

To improve reliability, a parallel estimation method is run using K-Means clustering on the depth pixel values within each bounding box. The depth pixels are grouped into clusters, where each cluster represents a distinct depth layer within the bounding box region. The number of clusters identified provides an independent estimate of the stack count that does not depend on physical calibration, instead relying on depth discontinuities as structural evidence of layering:

$$\text{stack_cluster} = \text{number of K-Means depth clusters within bounding box} \quad (4.4)$$

The two estimates are fused using a conservative maximum selection strategy to minimise the risk of undercounting:

$$\text{final_stack} = \max(\text{stack_gap}, \text{stack_cluster}) \quad (4.5)$$

A confidence score is assigned to each product based on how closely the two methods agree. If both produce the same stack count, confidence is rated High. A difference of one unit yields a Medium rating, and a difference of more than one unit yields Low confidence. This score is surfaced in the Web UI alongside the inventory figures so that operators can prioritise manual checks for products with low-confidence estimates. The final output of this module is a per-product stack count, a total estimated inventory figure (front count combined with hidden units), and the associated confidence level.

4.4 RESTOCK AND THRESHOLD MONITORING

The Restock and Threshold Monitoring module closes the loop between automated inventory counting and store operations. It takes the final per-product inventory count from Module 3 and evaluates each product against a predefined

minimum stock threshold. These thresholds are configured by store management during system setup and can be set independently for each product class to reflect different demand profiles, shelf sizes, or restocking lead times.

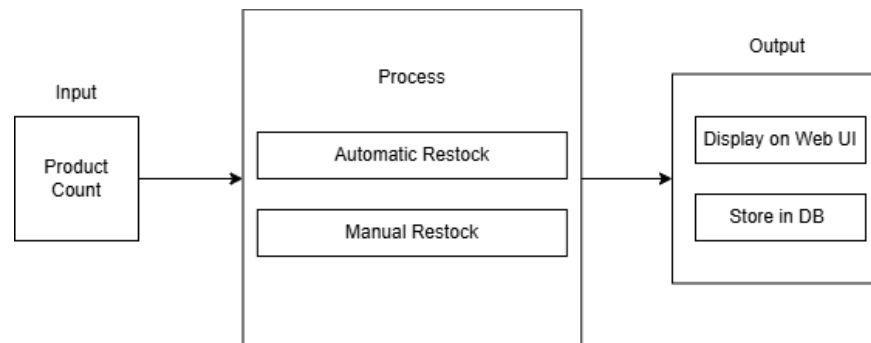


Figure 4.4: Restock and Threshold Monitoring Module: Threshold Evaluation and Restock Workflow

For each product, the estimated inventory count is compared against its corresponding threshold. If the count exceeds the threshold, the product’s status is recorded as Normal and no action is required. If the count falls at or below the threshold, the product is flagged and a restock record is automatically generated. This record is written to the database and includes the product identifier, current estimated count, threshold value, required restock quantity to reach the optimal stock level, and the timestamp of the detection event. In environments where the system is integrated with supply chain management software, this database record can directly trigger procurement or warehouse dispatch workflows.

Alongside automatic restock generation, flagged products are surfaced in the Web UI’s restock management panel, where store staff can review each alert and take manual action. Operators may confirm or dismiss an automatic restock order or manually initiate a restock for products that have not yet been re-scanned after replenishment. This manual override pathway is essential for handling edge cases such as recently restocked shelves, products temporarily out of position, or any scenario where the automated confidence score is low. All manual actions are logged with the operator’s identity and timestamp to ensure full auditability. The combined automatic and manual workflow ensures that the system remains robust and operationally trusted even in environments where detection accuracy is not perfect.

4.5 SUMMARY

This chapter described the four core modules of the smart retail inventory management system. Module 1 ensures that only clean, occlusion-free shelf images are captured by applying person detection and temporal presence logic to the incoming video stream. Module 2 uses transfer learning-based dense object detection to identify all front-visible products and produce a structured class-wise count. Module 3 extends this count into three dimensions by combining monocular depth estimation, empirical calibration, and K-Means clustering to estimate hidden stacked inventory, with confidence scoring to flag uncertain results. Module 4 applies rule-based threshold monitoring to the final inventory figures and generates both automatic and manually reviewable restock records. Together, these modules form a complete pipeline from raw video to actionable inventory management, with each stage designed to address a specific challenge in the retail shelf monitoring problem.

Chapter 5

IMPLEMENTATION

This chapter details the technical implementation of the AI-driven vision-based inventory management system. It outlines the core algorithms and logical procedures utilized to transition from raw video input to actionable, quantified inventory data. The system's execution is divided into four primary programmatic modules, each responsible for handling specific operational challenges: human occlusion, dense product detection, monocular depth estimation, and automated restocking logic.

5.1 SMART VISION CAPTURE ALGORITHM

To ensure that the deep learning models are not subjected to noise or occlusion caused by human shoppers obstructing the shelf view, the system employs a temporal logic algorithm for frame extraction. The system continuously monitors a predefined Region of Interest (ROI) and triggers a frame capture exclusively when the shelf has remained clear for a continuous, predefined temporal window (e.g., 3 seconds).

Algorithm 1 Human Presence Check and Clear Frame Capture

```
1: Input: Continuous Live Video Stream ( $V$ ), Shelf Region of Interest ( $ROI$ ),  
   Temporal Threshold ( $T_{clear} = 3s$ )  
2: Output: High-resolution, occlusion-free captured frame ( $F_{capture}$ )  
3: Initialize  $clear\_timer \leftarrow 0$   
4: while video stream  $V$  is active do  
5:    $current\_frame \leftarrow V.read\_frame()$   
6:    $detections \leftarrow PersonDetectionModel.predict(current\_frame)$   
7:    $occlusion\_detected \leftarrow False$   
8:   for each  $bounding\_box$  in  $detections$  do  
9:     if  $bounding\_box.class == \text{"person"}$  and  $bounding\_box \cap ROI \neq \emptyset$  then  
10:       $occlusion\_detected \leftarrow True$   
11:      break  
12:     end if  
13:   end for  
14:   if  $occlusion\_detected == True$  then  
15:      $clear\_timer \leftarrow 0$   
16:   else  
17:      $clear\_timer \leftarrow clear\_timer + \Delta t$   
18:   end if  
19:   if  $clear\_timer \geq T_{clear}$  then  
20:     return  $current\_frame$   
21:   end if  
22: end while
```

5.2 DENSE SHELF DETECTION AND NMS TUNING

ALGORITHM

In dense retail environments, products are placed in tight proximity, resulting in highly overlapping bounding boxes during the initial inference phase. Standard object detection pipelines frequently suppress these overlapping boxes, erroneously classifying them as duplicate detections. This implementation utilizes a tuned Non-Maximum Suppression (NMS) algorithm featuring a highly calibrated Intersection over Union (IoU) threshold to preserve valid front-facing products.

Algorithm 2 Dense Product Front Counting via Tuned NMS

```
1: Input: Captured Image ( $F_{capture}$ ), Fine-Tuned YOLOv8 Model ( $M_{YOLO}$ ),  
   Confidence Threshold ( $C_{th}$ ), IoU Threshold ( $I_{th} = 0.6$ )  
2: Output: Aggregated Front Product Count Mapping per SKU  
3:  $raw\_predictions \leftarrow M_{YOLO}.predict(F_{capture})$   
4:  $filtered\_boxes \leftarrow \emptyset$   
5: Step 1: Confidence Filtering  
6: for each  $prediction$  in  $raw\_predictions$  do  
7:   if  $prediction.confidence \geq C_{th}$  then  
8:      $filtered\_boxes.append(prediction)$   
9:   end if  
10: end for  
11: Step 2: Non-Maximum Suppression (NMS) Application  
12:  $final\_boxes \leftarrow Apply\_NMS(filtered\_boxes, I_{th})$   
13: Step 3: SKU Aggregation  
14:  $front\_inventory\_map \leftarrow HashMap()$   
15: for each  $box$  in  $final\_boxes$  do  
16:    $SKU \leftarrow box.class\_label$   
17:    $front\_inventory\_map[SKU] \leftarrow front\_inventory\_map[SKU] + 1$   
18: end for  
19: return  $front\_inventory\_map$ 
```

5.3 MONOCULAR DEPTH ESTIMATION ALGORITHM

To transcend the spatial limitations of 2D vision systems, the architecture utilizes the Depth Anything V2 model to estimate the volume of inventory concealed behind the front-facing units. This algorithmic procedure converts extracted relative depth map values into physical metric estimates via a calibrated scaling factor.

Algorithm 3 Depth-Based Stack Estimation

- 1: **Input:** Product Bounding Box (B), Generated Depth Map (D_{map}), Known Physical Product Depth (P_{depth_cm}), Depth Scale Factor (S_{factor})
 - 2: **Output:** Estimated Stack Count ($Total_Stack$)
 - 3: **Step 1: Depth Feature Extraction**
 - 4: $front_depth_rel \leftarrow Median(D_{map} \cap B)$
 - 5: $wall_depth_rel \leftarrow Median(D_{map} \cap Reference_Strip_Above(B))$
 - 6: **Step 2: Spatial Gap Calculation**
 - 7: $depth_gap_rel \leftarrow |front_depth_rel - wall_depth_rel|$
 - 8: $physical_gap_cm \leftarrow depth_gap_rel \times S_{factor}$
 - 9: **Step 3: Inventory Approximation**
 - 10: $estimated_hidden_units \leftarrow \lfloor \frac{physical_gap_cm}{P_{depth_cm}} \rfloor$
 - 11: $Total_Stack \leftarrow 1 + estimated_hidden_units$
 - 12: **return** $Total_Stack$
-

5.4 AUTOMATED THRESHOLD MONITORING LOGIC

The final stage of the implementation integrates the generated 2D front count and 3D stack estimates to execute automated operational decisions. The algorithm evaluates the holistic inventory state against predefined minimum thresholds, subsequently triggering database updates and user interface alerts.

Algorithm 4 Automated Restock Decision Engine

```
1: Input:   Computed Front Count ( $Count_{front}$ ), Computed Stack Count  
             ( $Count_{stack}$ ), Product Identifier ( $SKU_{ID}$ ), System Database ( $DB$ )  
2: Output: Operational Alert Status and Persisted Database State  
3:  $Total\_Inventory \leftarrow Count_{front} + Count_{stack}$   
4:  $Min\_Threshold \leftarrow DB.fetch\_threshold(SKU_{ID})$   
5:  $Optimal\_Stock \leftarrow DB.fetch\_optimal\_capacity(SKU_{ID})$   
6: if  $Total\_Inventory \leq Min\_Threshold$  then  
7:    $Alert\_Status \leftarrow \text{"RESTOCK\_REQUIRED"}$   
8:    $Restock\_Amount \leftarrow Optimal\_Stock - Total\_Inventory$   
9:    $DB.log\_restock\_event(SKU_{ID}, Total\_Inventory, Restock\_Amount, Current\_Timestamp)$   
10:   $UI.trigger\_alert(SKU_{ID}, Alert\_Status)$   
11: else  
12:   $Alert\_Status \leftarrow \text{"NORMAL"}$   
13:   $DB.update\_inventory\_state(SKU_{ID}, Total\_Inventory, Current\_Timestamp)$   
14: end if  
15: return  $Alert\_Status$ 
```

Chapter 6

RESULTS AND PERFORMANCE ANALYSIS

This chapter evaluates the implemented AI-driven vision-based inventory management system. The evaluation is divided into a qualitative assessment of the system's operational outputs through the user interface and a quantitative performance analysis of the underlying deep learning models.

6.1 RESULTS AND INTERFACES

This section showcases the system's operational capabilities, demonstrating how raw video feeds are transformed into actionable inventory data through intelligent capture, detection, and threshold monitoring.

6.1.1 SMART VISION CAPTURE: SOLVING HUMAN OCCLUSION

A major limitation of traditional vision systems is their vulnerability to occlusion caused by shoppers. The Smart Vision Capture module effectively resolves this by utilizing human presence detection to trigger image captures only when the shelf is clear.

- **Before Image:** Demonstrates the live video feed with a customer obstructing the camera's view of the shelf. The system actively suppresses capture during this phase.
- **After Image:** Shows the clean, unobstructed high-resolution frame captured by the system after the 3-second clearance window, ensuring the downstream detection models receive optimal data.

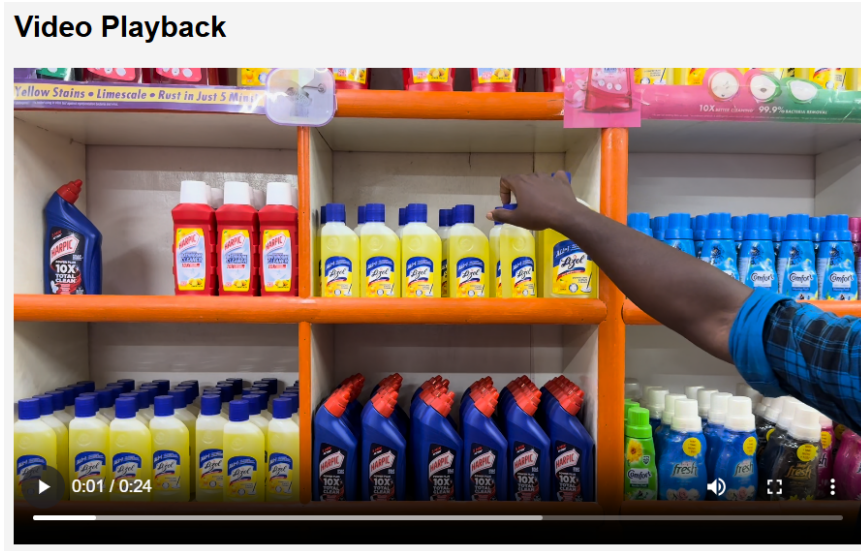


Figure 6.1: Smart Vision Capture successfully avoiding human occlusion.

6.1.2 ANNOTATED DETECTION OUTPUT

Upon processing the clear shelf image, the fine-tuned YOLOv8 model identifies the visible front-facing products. The output image demonstrates the model's precision, rendering tight bounding boxes around tightly packed items. Each bounding box is overlaid with the predicted product class label and a confidence score, visually confirming the system's ability to handle dense retail environments.



Figure 6.2: YOLOv8 annotated output showing bounding boxes and classifications.

6.1.3 PRODUCT COUNTED DASHBOARD

The processed data is aggregated and visualized on the web-based inventory dashboard. This interface translates the 2D bounding boxes and 3D depth estimations into a comprehensive inventory summary. The dashboard provides store operators with a clear tabular view of the current stock levels, displaying the calculated front count alongside the estimated total inventory for each specific SKU.

Final Inventory Summary			
Product	Front (M2)	Stack (M3)	Total Count
Comfort	8	13	21
Fresh	7	14	21
Lizol	14	26	40
Harpic Red	3	6	9
Harpic	7	14	21
Grand Total: 112			

Figure 6.3: Web dashboard displaying real-time aggregated product counts.

6.1.4 RESTOCK ALERT SYSTEM

To bridge the gap between passive monitoring and active operational management, the system evaluates the final product counts against predefined minimum thresholds. When a product's stock level falls below this threshold, the system automatically triggers a visual restock alert within the interface, highlighting the specific item and recommending a replenishment action to store staff.

Restock Dashboard			
Product	Quantity	Status	Action
Pepsi	2	LOW	<button>Restock</button>
Fanta	8	OK	-
Coke	3	LOW	<button>Restock</button>

Figure 6.4: Automated restock alert generated for low-inventory items.

6.2 PERFORMANCE ANALYSIS

This section provides a quantitative evaluation of the YOLOv8 model, confirming its accuracy and reliability for commercial deployment.

The object detection model was fine-tuned on the custom product dataset. The training phase demonstrated rapid convergence and high learning stability, evaluated across three primary dimensions: localization loss, evaluation metrics, and classification accuracy.

6.2.1 TRAINING AND VALIDATION LOSS

The training and validation box loss showed a consistent downward trend across the epochs. This steady decrease indicates that the model successfully learned tight bounding box localization and generalized well to unseen validation data without experiencing severe overfitting. Proper box localization is essential for accurately framing individual items on dense retail shelves.

Table 6.1: YOLOv8 Training and Validation Loss Over Epochs

Epoch	Train Box Loss	Train Class Loss	Val Box Loss	Val Class Loss
1	1.8478	1.8478	1.9606	1.8478
2	1.7807	1.8478	1.6806	1.8478
⋮	⋮	⋮	⋮	⋮
9	1.2303	1.8478	1.2567	1.8478
10	1.1728	1.8478	1.2250	1.8478

6.2.2 EVALUATION METRICS (PRECISION AND mAP)

The model achieved high precision and mean Average Precision (mAP), proving its robustness in distinguishing between similar product packaging and handling edge boundaries. High precision ensures that the model correctly identifies actual products rather than background artifacts, while a strong mAP score reflects the model's overall detection reliability across various confidence thresholds.

Table 6.2: YOLOv8 Precision, Recall and mAP Metrics Over Epochs

Epoch	Precision	Recall	mAP50	mAP50-95
1	1.8478	1.8478	1.9606	1.8478
2	1.7807	1.8478	1.6806	1.8478
⋮	⋮	⋮	⋮	⋮
9	1.2303	1.8478	1.2567	1.8478
10	1.1728	1.8478	1.2250	1.8478

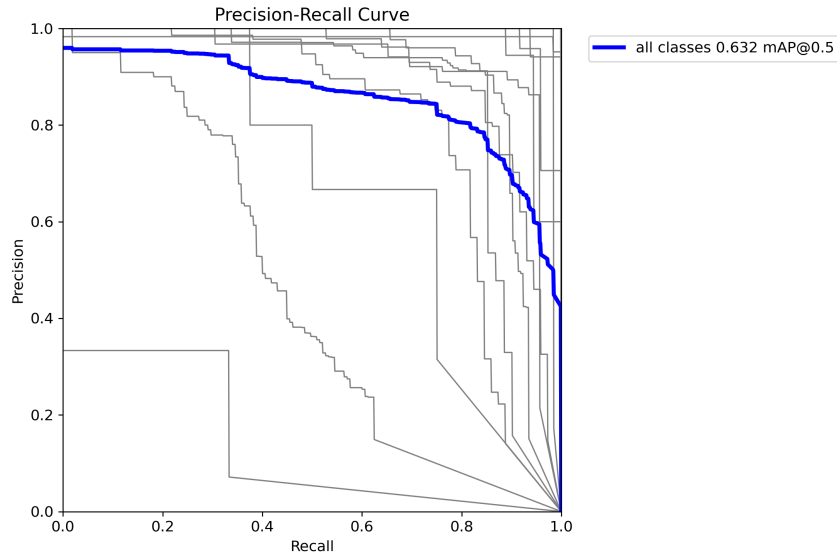


Figure 6.5: Precision and mean Average Precision (mAP) evaluation metrics.

6.2.3 CONFUSION MATRIX AND CLASSIFICATION ACCURACY

The normalized confusion matrix highlights the model's classification accuracy. It shows high true positive rates across the trained product classes and minimal confusion with background elements. This ensures that false detections on empty shelf space are kept to an absolute minimum, which is a critical requirement for maintaining an accurate front-row inventory count.

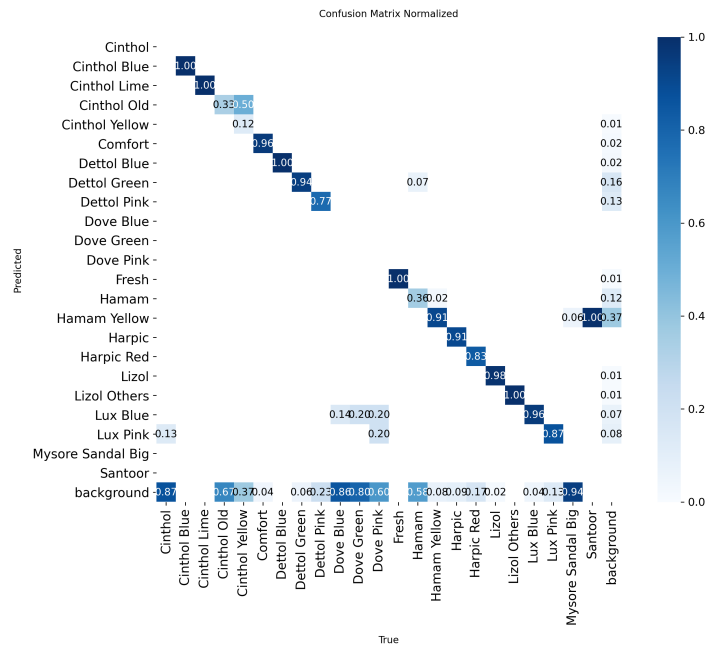


Figure 6.6: Normalized confusion matrix demonstrating class-level accuracy and minimal background confusion.

Chapter 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This project successfully designed and implemented an AI-driven, vision-based inventory management system capable of automating retail shelf monitoring and mitigating out-of-stock scenarios. Traditional inventory practices, which rely heavily on manual audits, are labor-intensive and prone to operational delays. The proposed architecture addresses these inefficiencies by establishing a continuous, intelligent pipeline that operates without physical human intervention.

The system's modular design proved highly effective in navigating the complexities of dense retail environments. The Smart Vision Capture module successfully mitigated human occlusion, ensuring that downstream models processed only clear shelf images. By utilizing a fine-tuned YOLOv8 model alongside the Depth Anything V2 model for 3D stack estimation, the system accurately calculated both visible and hidden inventory. Finally, the integration of automated threshold logic and a web-based dashboard successfully translated computer vision inferences into actionable restocking alerts, significantly enhancing retail efficiency and reducing the manual workload for store operators.

7.2 FUTURE WORK

While the current system demonstrates strong real-time performance, several avenues exist to further improve its scalability and commercial viability:

- **Integration with Point of Sale (POS) Systems:** By linking the vision-based inventory counts with real-time checkout data from the store's POS system, the architecture could automatically detect discrepancies. This would help identify misplaced items (planogram compliance errors) or instances of store shrinkage (theft).
- **Predictive Analytics for Demand Forecasting:** Integrating Long Short-Term Memory (LSTM) networks or time-series forecasting models to analyze historical depletion rates. By learning when specific products sell out (based on seasonality or time of day), the system could shift from reactive alerts to predictive

restocking, ordering inventory before the shelf even drops below the threshold.

- **Multi-Camera Aisle Stitching:** The current system operates on a segmented shelf view. Future iterations could stitch video feeds from multiple cameras along an entire aisle to create a continuous, panoramic inventory map, allowing for large-scale, store-wide monitoring.
- **Mobile Application for Store Staff:** Transitioning the Web UI into a dedicated mobile application for retail employees. This app could push real-time, location-specific restocking alerts directly to a worker's smartphone or smartwatch, further reducing response times.
- **Hardware Upgrade to Stereo Depth Sensors:** While the monocular Depth Anything V2 model performs well for relative stack estimation, integrating cost-effective RGB-D (stereo) cameras could provide absolute metric depth. This hardware enhancement would yield millimeter-accurate gap calculations without the need for reference object calibration.

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